

Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY NA.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE85,000 × 100,000 mm

BOARD THICKNESS1,602 mm

TRACE WIDTH0,300 mm

TRACE TO TRACE-0,000 mm

MIN. HOLE (PTH)0,250 mm

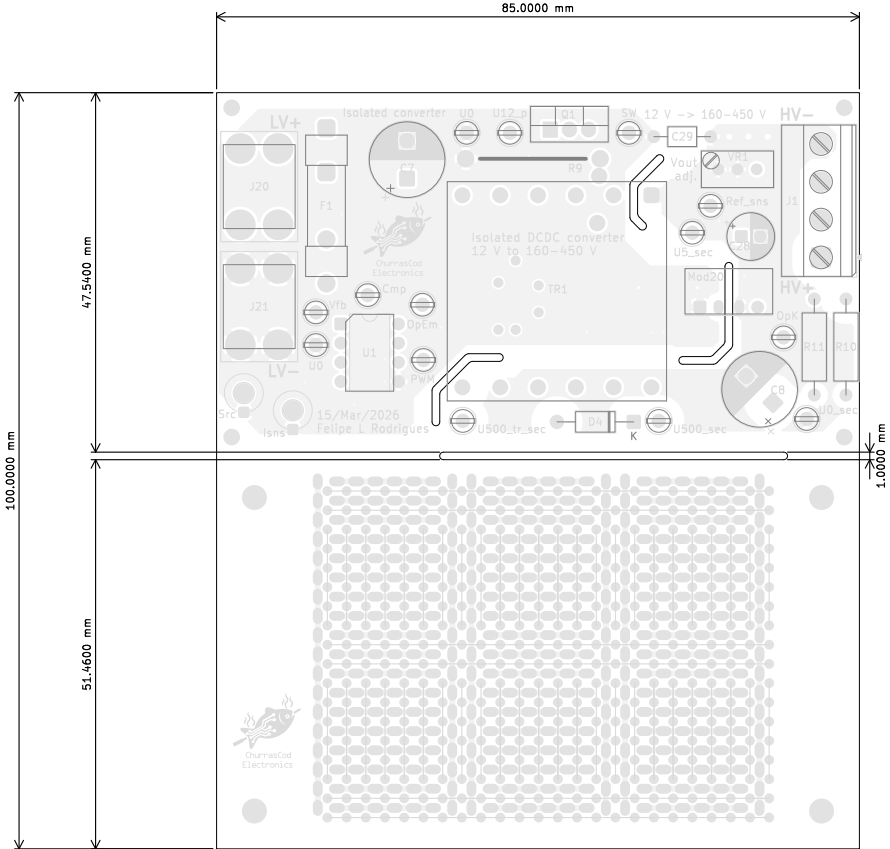
MIN. HOLE (NPTH)2,100 mm

ANNULAR RING0,150 mm

COPPER TO HOLE0,200 mm

COPPER TO EDGE0,250 mm

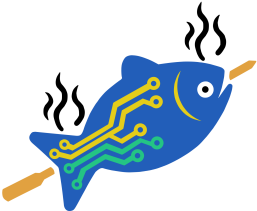
HOLE TO HOLE0,254 mm



Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen			Direct Printing	Legend	GBR
F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
Core		1,48mm	FR4_7628	Dielectric	
Copper	L2 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating



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Comments:

Company:

NA

Variant:

CHECKED

Git Hash:

9fc4300

Board Name:

Step-Up

Project Name:

Isolated step-up 12:450 V

Sheet Title:

Top Fabrication (Scale 1:1)

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Date:

2026-02-07

Revision:

2.0.0+ (Unreleased)

Sheet Path:

Reviewer:

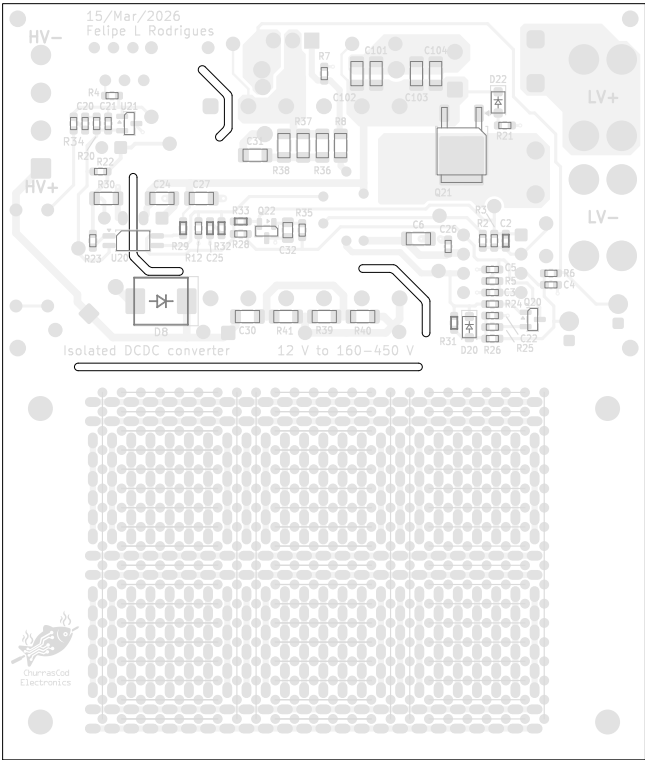
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Size:

A4

Sheet:

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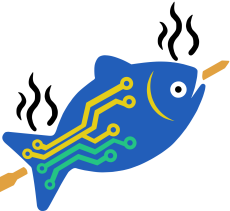
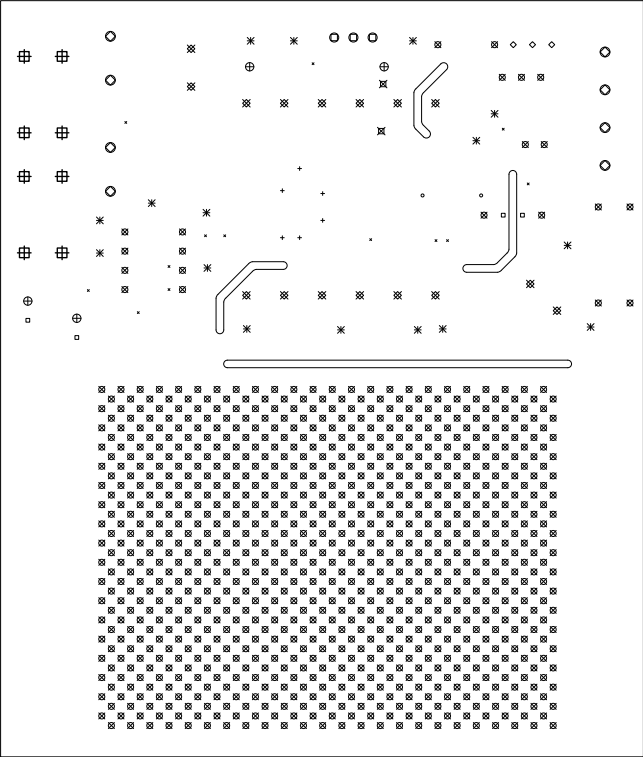
Comments:	Company: NA		Variant: CHECKED	Git Hash: 9fc4300
	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 2 of 7

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0.25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0.40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
+	6	0.50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
□	4	0.70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	3	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	885	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
✱	16	0.90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊗	2	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	16	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	4	1.10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	3	1.20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
○	8	1.30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	8	1.80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
Total 970						



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Isolated step-up 12:450 V

Sheet Title:

Drill Drawing (L1 - L4)

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Isolated_12to450V.kicad_pcb

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FR

Size:

A4

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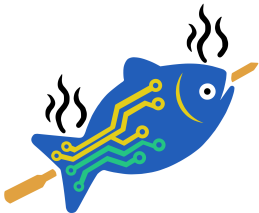
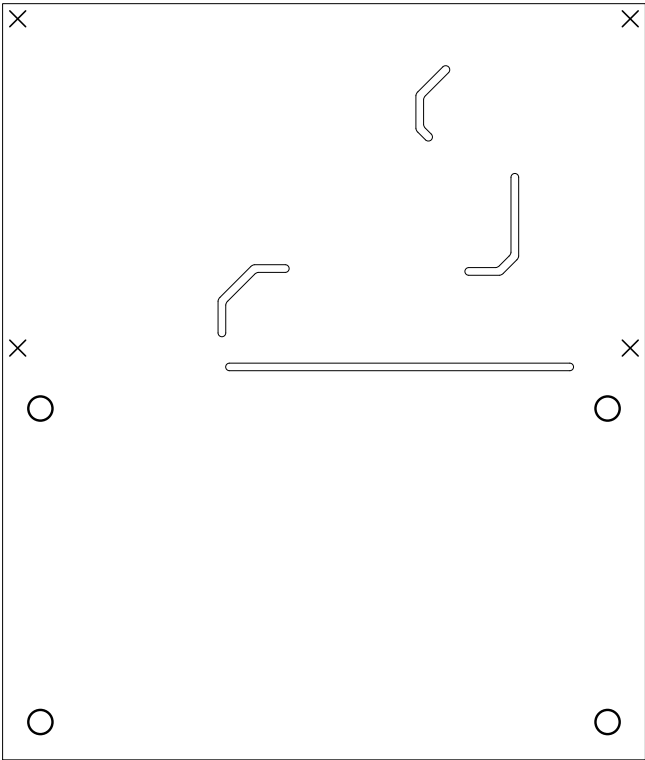
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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
○	4	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
Total 8						



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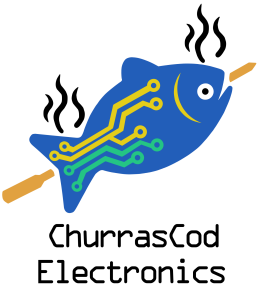
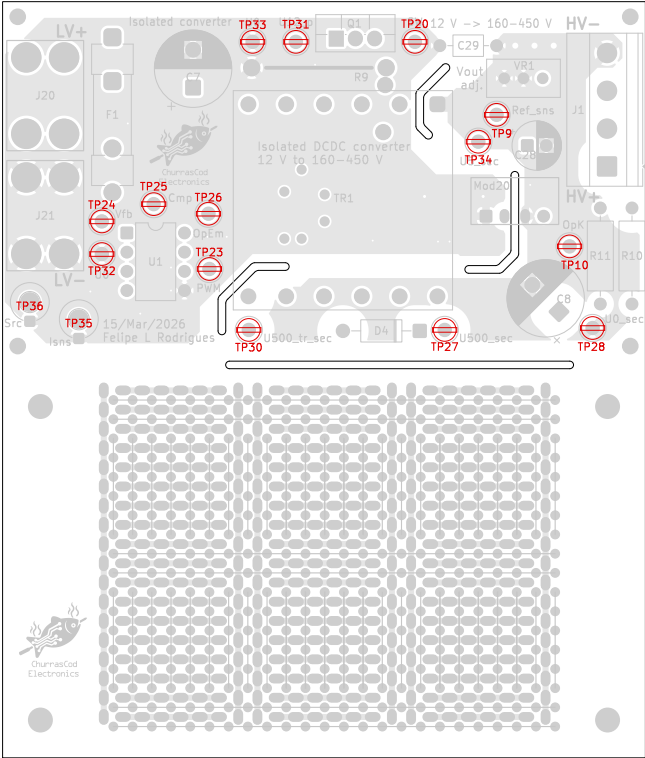
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	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V		
Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)	
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 4 of 7	

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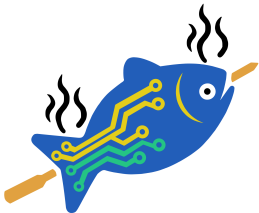
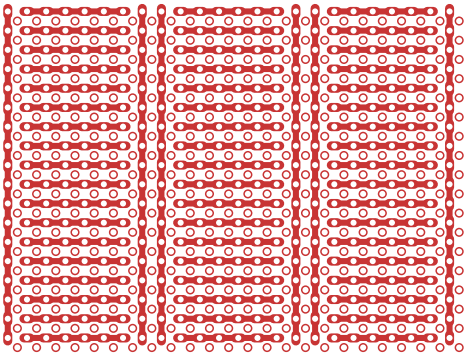
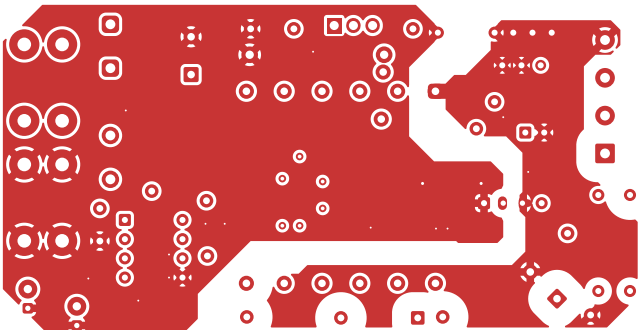
Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP9	Sec_ref_sns	67.32	35.01
TP10	Sec_OpK	76.97	17.61
TP20	SW	56.53	44.66
TP23	PWM	29.35	14.62
TP24	Vfb	15.13	20.91
TP25	Comp	21.98	23.20
TP26	OpEm	29.22	21.93
TP27	U500_sec	60.46	6.56
TP28	U0_sec	80.02	6.81
TP30	U500_tr_sec	34.56	6.56
TP31	U12_p	40.78	44.66
TP32	U0	15.13	16.59
TP33	U0	35.06	44.66
TP34	U5_sec	64.91	31.45
TP35	Isns,U0	12.08	6.70
TP36	Src	5.60	8.97
TP102	Net-(TP101-Pad1)	23.00	-1.44
TP103	Net-(TP101-Pad1)	23.00	-1.44
TP104	Net-(TP101-Pad1)	25.54	-1.44
TP105	Net-(TP101-Pad1)	28.08	-1.44
TP106	Net-(TP101-Pad1)	30.62	-1.44
TP107	Net-(TP107-Pad1)	17.92	-3.98
TP108	Net-(TP107-Pad1)	20.46	-3.98
TP109	Net-(TP107-Pad1)	23.00	-3.98
TP110	Net-(TP107-Pad1)	25.54	-3.98
TP111	Net-(TP107-Pad1)	28.08	-3.98
TP112	Net-(TP107-Pad1)	30.62	-3.98
TP113	Net-(TP113-Pad1)	17.92	-6.52
TP114	Net-(TP113-Pad1)	20.46	-6.52
TP115	Net-(TP113-Pad1)	23.00	-6.52
TP116	Net-(TP113-Pad1)	25.54	-6.52

All dimensions are in millimeters
unless otherwise specified.



Comments:	Company:		Variant:	Git Hash:
	NA		CHECKED	9fc4300
	Board Name:		Project Name:	
Sheet Title:	Step-Up		Isolated step-up 12:450 V	
	Top Test Points (Scale 1:1)		Date:	Revision:
	Isolated_12to450V.kicad_pcb		2026-02-07	2.0.0+ (Unreleased)
Sheet Path:		Designer:	Size:	Sheet:
		FR	A4	5 of 7
		Reviewer:		
		FR		

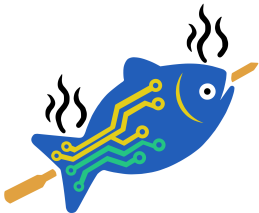
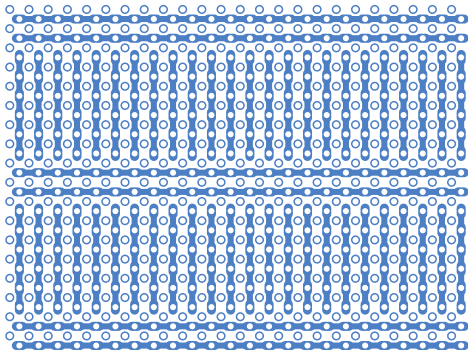
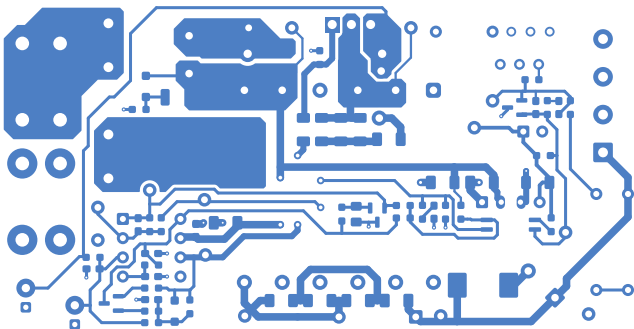


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Comments:	Company:		Variant:	Git Hash:
	NA		CHECKED	9fc4300
	Board Name:		Project Name:	
Sheet Title:	File Name:	Designer:	Date:	Revision:
F.Cu (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-07	2.0.0+ (Unreleased)
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Step-Up Fabrication Document

B.Cu (Scale 1:1)



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	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V				
	Sheet Title: B.Cu (Scale 1:1)		File Name: Isolated_12to450V.kicad_pcb		Designer: FR		
		Date:		Revision:			
		2026-02-07		2.0.0+ (Unreleased)			
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